MX

In [1]:
```
from PT3S import Mx
```

In [2]:
```
mx=Mx.Mx(r'..\testdata\M-1-0-1.MX1')
```

with Pandas

MX-Channel TIMESTAMP (1st MX-Channel) used as DataFrame Index; all other MX-Channels are DataFrame Columns

In [3]:
```
import pandas as pd
```

DataFrame

In [4]:
```
df=mx.getPdfFromMxsZipFile(mxsZipFile=r'..\testdata\M-1-0-1.ZIP')
type(df)
```

Out[4]:
```
pandas.core.frame.DataFrame
```

Series

In [5]:
```
ts=df.iloc[1:-2,0]
type(ts)
```

Out[5]:
```
pandas.core.series.Series
```

Timestamps

In [6]:
```
ts.index[0]
```

Out[6]:
```
Timestamp('2017-06-18 23:00:00')
```

In [7]:
```
ts.index[-1]
```

Out[7]:
```
Timestamp('2017-06-19 00:30:00')
```

DatetimeIndex

In [8]:
```
type(ts.index)
```

Out[8]:
```
pandas.tseries.index.DatetimeIndex
```

Time Zone (tz_convert(None) will remove timezone after converting to UTC time)

In [9]:
```
print(ts.index.tz)
```

```
None
```

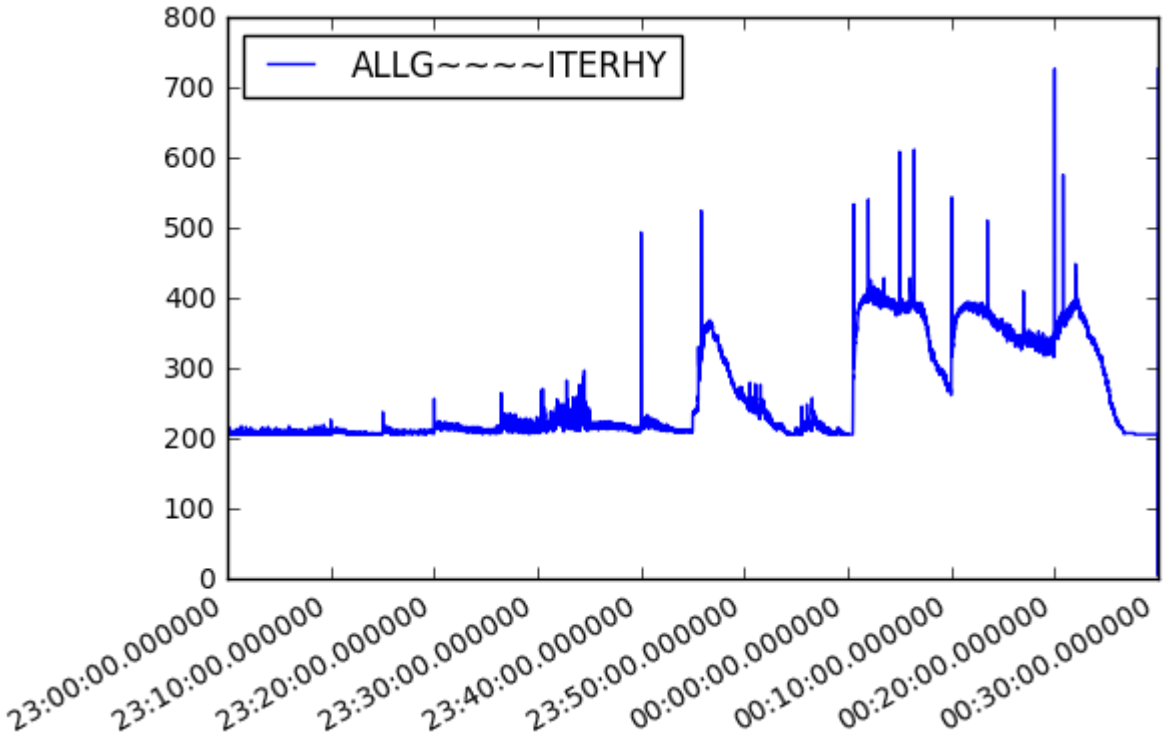
Time Series Plot

In [10]:
```
%matplotlib inline
import matplotlib.pyplot as plt
from matplotlib import dates
```

by MX-Channel-Idx

In [11]:
```
plt.figure()
df.iloc[:,17].plot()
plt.legend(loc='best')
```

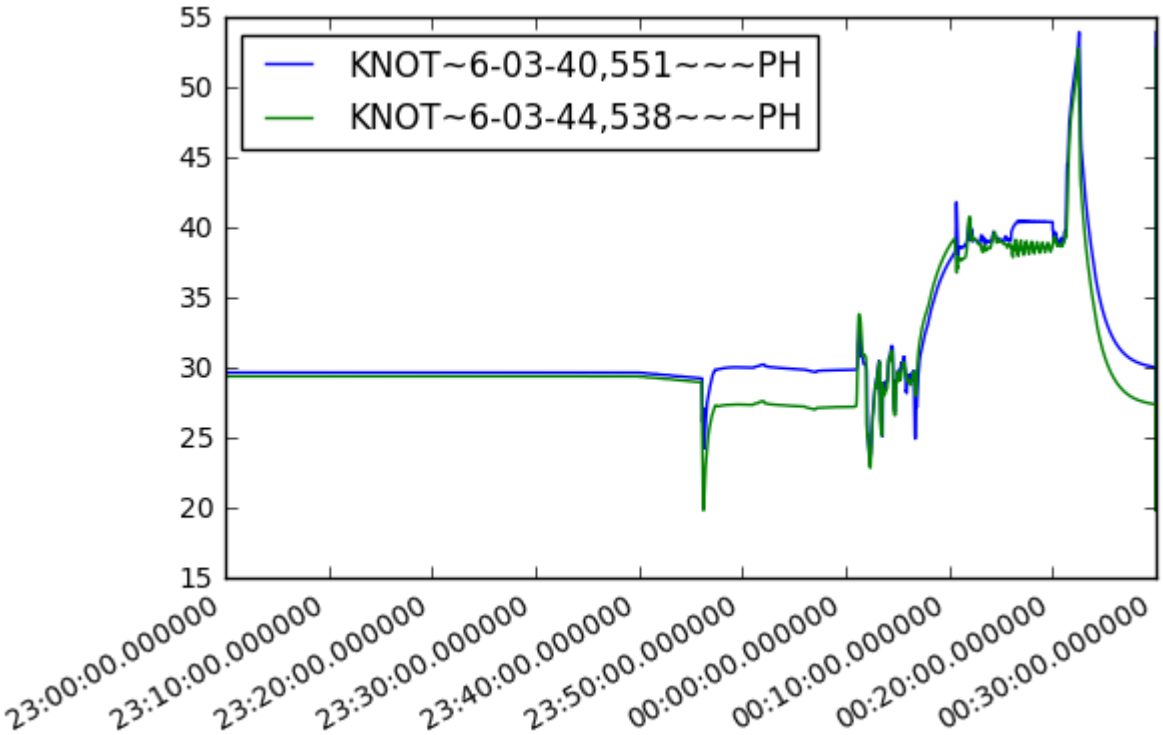
Out[11]: <matplotlib.legend.Legend at 0x1fdf51b4518>



by MX-Channel-Sir3sID

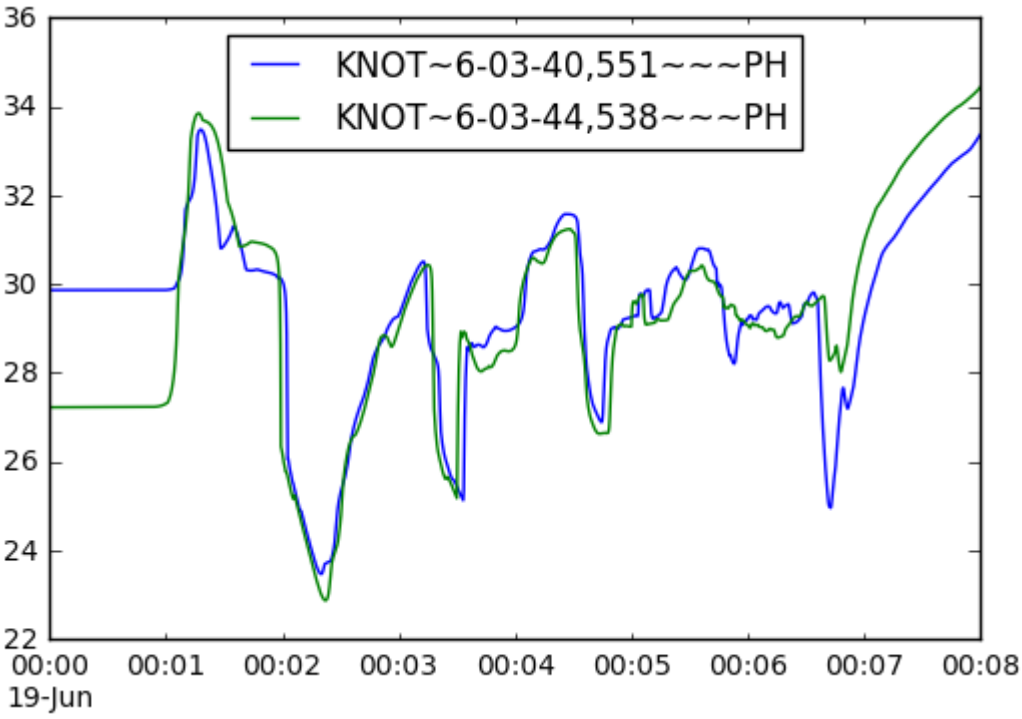
In [12]:
```
import re
regExpCompiled=re.compile(r'^KNOT~6-03-4\d+(\S+)~(\S*)~(\S*)~PH')
df_colsel=df.loc[:,sorted([column for column in df.columns if regExpCompiled.search(c
olumn) != None])]
df_colsel.plot()
```

Out[12]: <matplotlib.axes._subplots.AxesSubplot at 0x1fdfb6dfeb8>



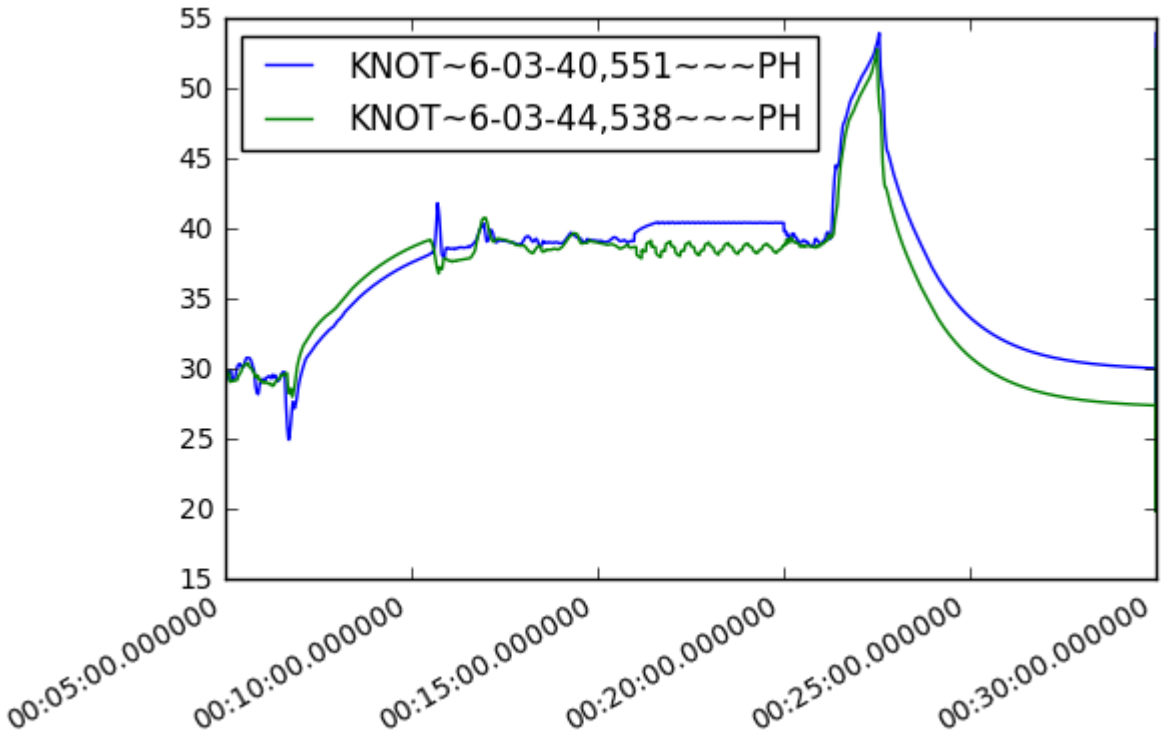
In [13]:
```
start=df_colsel.index[0].to_datetime()+pd.DateOffset(minutes=60)
end=start+pd.DateOffset(minutes=8)
df_colsel.between_time(start_time=start.time(), end_time=end.time()).plot()
```

Out[13]: <matplotlib.axes._subplots.AxesSubplot at 0x1fd92642128>



In [14]:
```
df_colsel[pd.datetime(2017,6,19,0,5):].plot()
```

Out[14]:
```
<matplotlib.axes._subplots.AxesSubplot at 0x1fd9261c9e8>
```



SnapShotType

In [15]:
```
len(df)
```

Out[15]:
```
21604
```

In [16]:
```
df.loc[:,[r'ALLG~~~~SNAPSHOTTYPE']].head(1)
```

Out[16]:

	ALLG~~~~SNAPSHOTTYPE
2017-06-18 23:00:00	b'STAT'

In [17]:
```
df.loc[:,[r'ALLG~~~~SNAPSHOTTYPE']].tail(2)
```

Out[17]:

	ALLG~~~~SNAPSHOTTYPE
2017-06-19 00:30:00	b'TMIN'
2017-06-19 00:30:00	b'TMAX'

In [18]:
```
len(df[df[r'ALLG~~~~SNAPSHOTTYPE'].isin([b'STAT',b'TMIN',b'TMAX'])])
```

Out[18]:
```
3
```

In [19]:
```
df.loc[:,
[r'ALLG~~~~SNAPSHOTTYPE',r'ALLG~~~~ITERHY']].groupby(r'ALLG~~~~SNAPSHOTTYPE').count()
```

Out[19]:

	ALLG~~~~ITERHY
ALLG~~~~SNAPSHOTTYPE	
b'STAT'	1
b'TIME'	21601
b'TMAX'	1
b'TMIN'	1

In [20]:
```
df2=df[df[r'ALLG~~~~SNAPSHOTTYPE'].isin([b'TIME'])]
```

In [21]:
```
len(df2)
```

Out[21]:
```
21601
```

In [22]:
```
df2.loc[:,'KNOT~6-03-44,538~~~PH'].head(4)
```

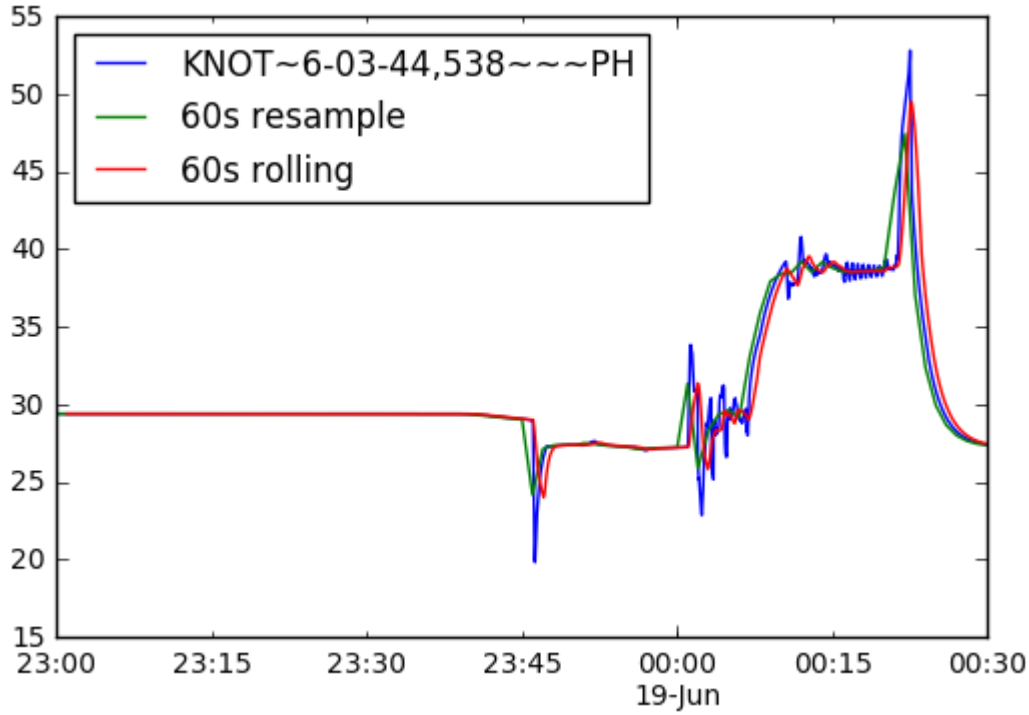
Out[22]:
```
2017-06-18 23:00:00.000    29.404100
2017-06-18 23:00:00.250    29.404099
2017-06-18 23:00:00.500    29.404100
2017-06-18 23:00:00.750    29.404100
Name: KNOT~6-03-44,538~~~PH, dtype: float64
```

Time Series Functions

In [23]:
```
ts=df2.loc[:,'KNOT~6-03-44,538~~~PH']
```

In [24]:
```
ts60=df2.loc[:,'KNOT~6-03-44,538~~~PH'].resample('60s').mean()
ts60.name='60s resample'
ts60r = ts.rolling(window=60*4).mean()
ts60r.name='60s rolling'
plt.figure()
ts.plot()
ts60.plot()
ts60r.plot()
plt.legend(loc='best')
```

Out[24]:
```
<matplotlib.legend.Legend at 0x1fdfbe2e978>
```



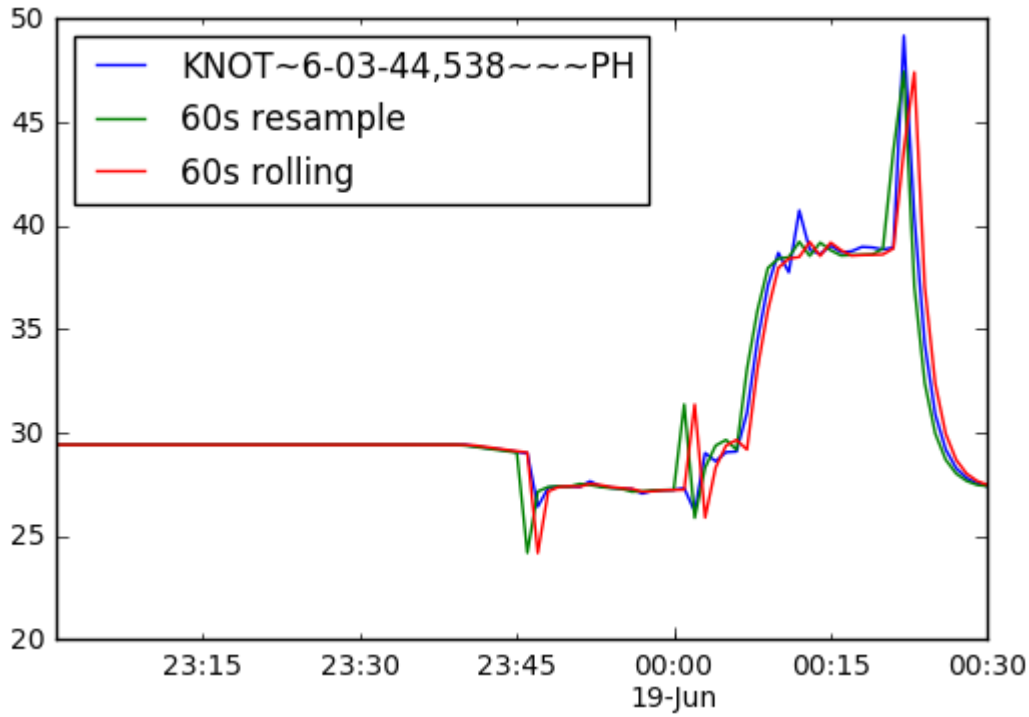
In [25]:
```
df3=pd.DataFrame({ts60.name : ts60,ts.name : ts,ts60r.name : ts60r})
df3=df3.reindex_axis([ts.name,ts60.name,ts60r.name], axis=1)
df3.head()
```

Out[25]:

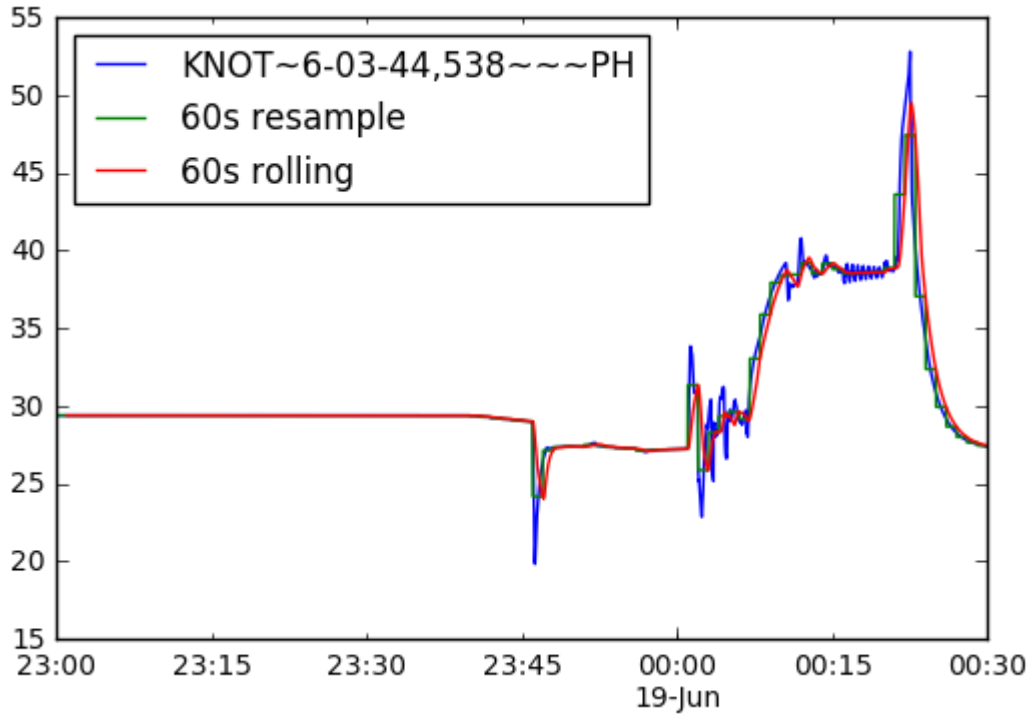
	KNOT~6-03-44,538~~~PH	60s resample	60s rolling
2017-06-18 23:00:00.000	29.404100	29.404088	NaN
2017-06-18 23:00:00.250	29.404099	NaN	NaN
2017-06-18 23:00:00.500	29.404100	NaN	NaN
2017-06-18 23:00:00.750	29.404100	NaN	NaN
2017-06-18 23:00:01.000	29.404099	NaN	NaN

In [26]:
```
df3.dropna().plot()
```

Out[26]: <matplotlib.axes._subplots.AxesSubplot at 0x1fd87a947f0>



In [27]:
```
pPlot = df3.ffill().plot()
# type(pPlot) # matplotlib.axes._subplots.AxesSubplot
```

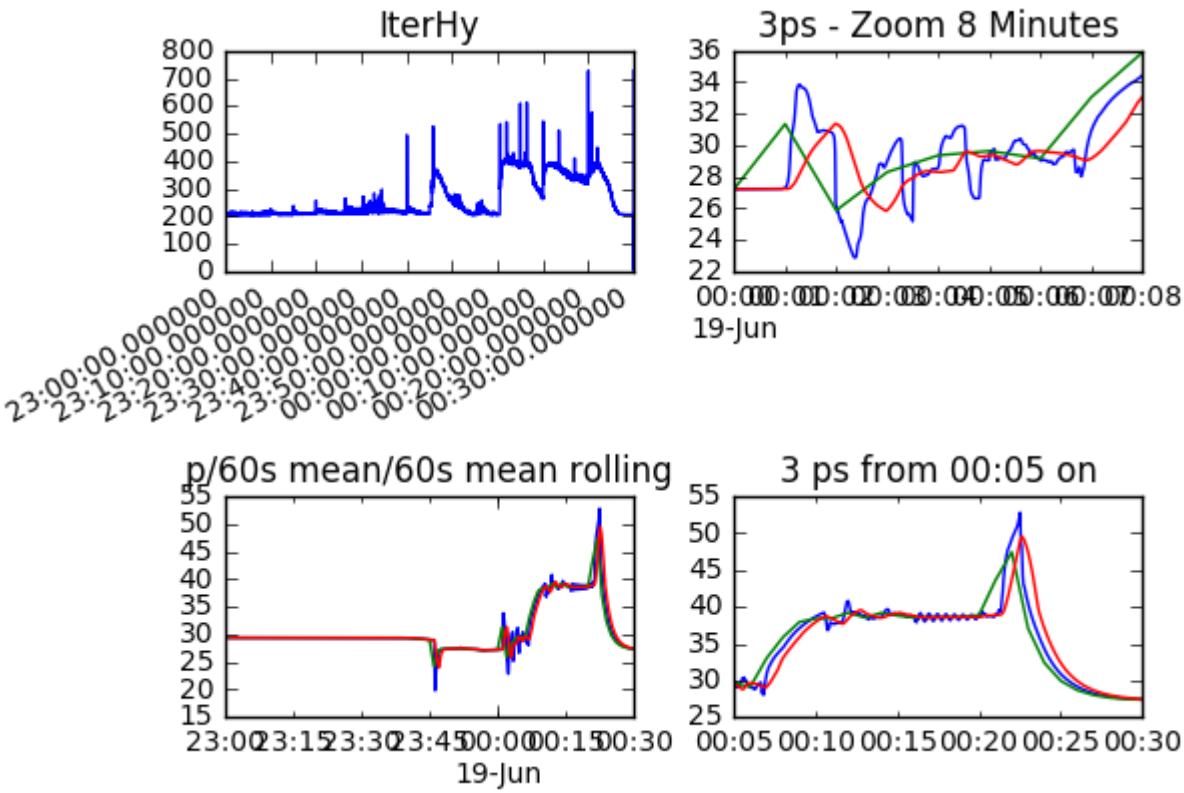


MATLAB, and pyplot, have the concept of the current figure and the current axes. All plotting commands apply to the current axes.

The function gca() returns the current axes (a matplotlib.axes.Axes instance), and gcf() returns the current figure (matplotlib.figure.Figure instance).

The subplot() command specifies numrows, numcols, num where num ranges from 1 to numrows*numcols.

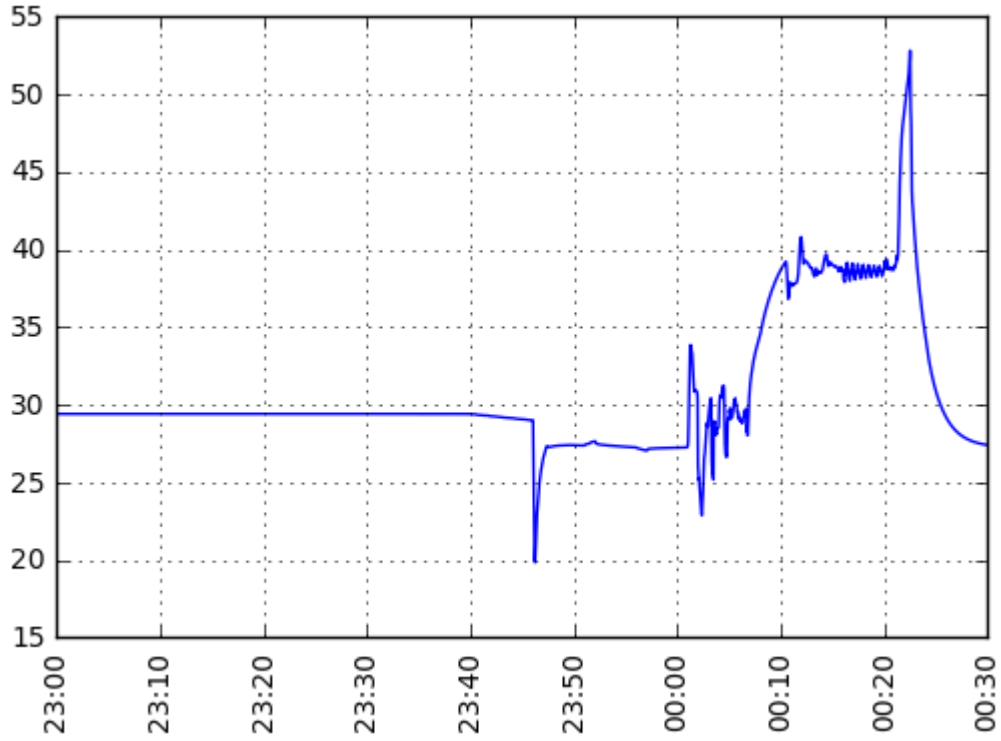
In [28]:
```
plt.figure(1)

plt.subplot(221)
plt.title('IterHy')
df.iloc[:,17].plot()

plt.subplot(223)
plt.title('p/60s mean/60s mean rolling')
ts.plot()
ts60.plot()
ts60r.plot()

plt.subplot(222)
plt.title('3ps - Zoom 8 Minutes')
ts.between_time(start_time=start.time(), end_time=end.time()).plot()
ts60.between_time(start_time=start.time(), end_time=end.time()).plot()
ts60r.between_time(start_time=start.time(), end_time=end.time()).plot()

plt.subplot(224)
plt.title('3 ps from 00:05 on')
ts[pd.datetime(2017,6,19,0,5):].plot()
ts60[pd.datetime(2017,6,19,0,5):].plot()
ts60r[pd.datetime(2017,6,19,0,5):].plot()
plt.tight_layout(pad=0.4, w_pad=0.5, h_pad=1.0) #plt.subplots_adjust(top=0.92, ...)

plt.show()
```



In [29]:
```
#fig, ((ax1, ax2), (ax3, ax4)) = plt.subplots(nrows=2, ncols=2)
#example_plot(ax1)
#def example_plot(ax, fontsize=12):
#    ax.plot(...)
```

"Hand-made" xaxis

In [30]:
```
plt.close('all')
plt.figure()
plt.subplot()
plt.gca().xaxis.set_major_locator(dates.MinuteLocator(interval=10))
plt.gca().xaxis.set_major_formatter(dates.DateFormatter('%H:%M'))
#plt.gcf().autofmt_xdate()
plt.plot(ts.index, ts)
labels=plt.gca().get_xticklabels()
plt.setp(labels,rotation=90)
plt.grid()
plt.show()
```



In [ ]: